ECA1804 - Optoelectronics and Optical Communication.

Exp no: 20 PERFORMANCE EVALUATION OF FREE-SPACE OPTICAL (FSO) LINK UNDER ATMOSPHERIC TURBULENCE USING OPTISYSTEM.

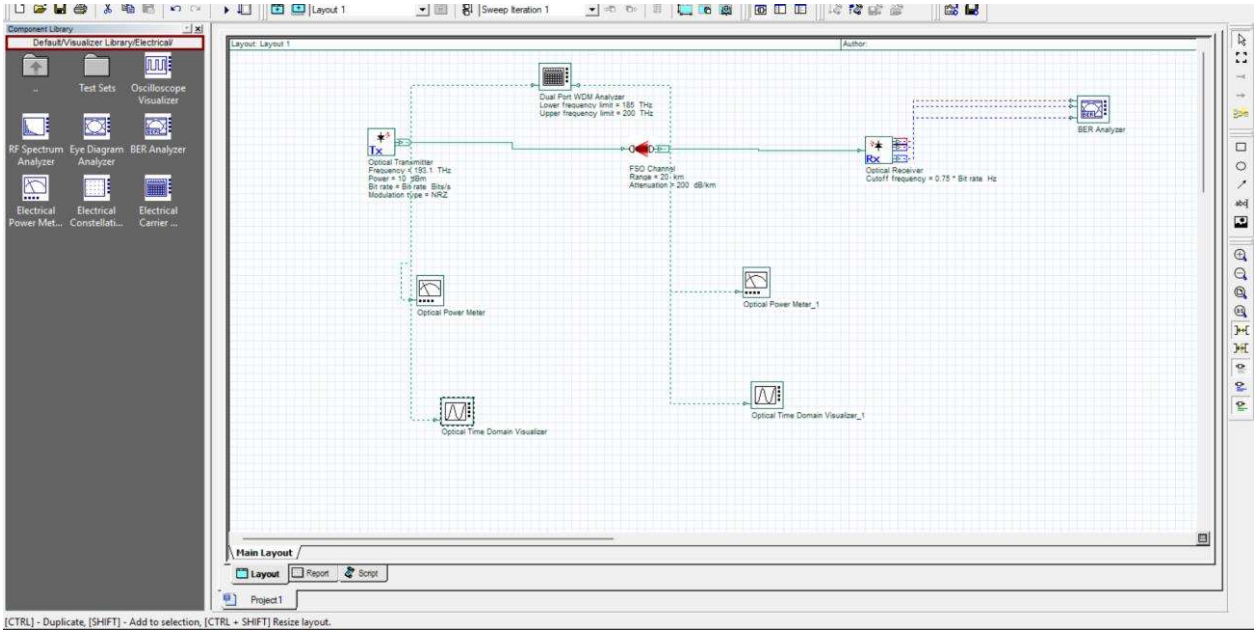
OBJECTIVES

1. Analyze the effect of range on received power, considering the combined contributions of geometric spreading loss and atmospheric attenuation.
2. Evaluate how the combined effects of increasing range and attenuation influence received power and bit error rate (BER).
3. Assess the extent to which increased transmit power can compensate for attenuation-induced signal loss.

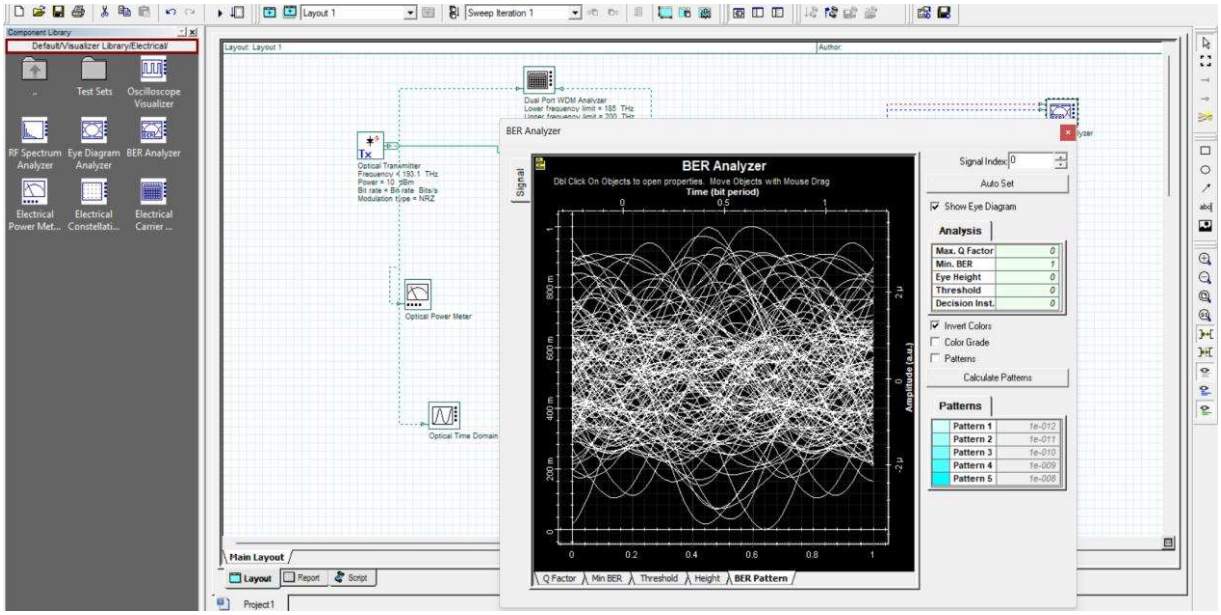
SOFTWARE REQUIRED

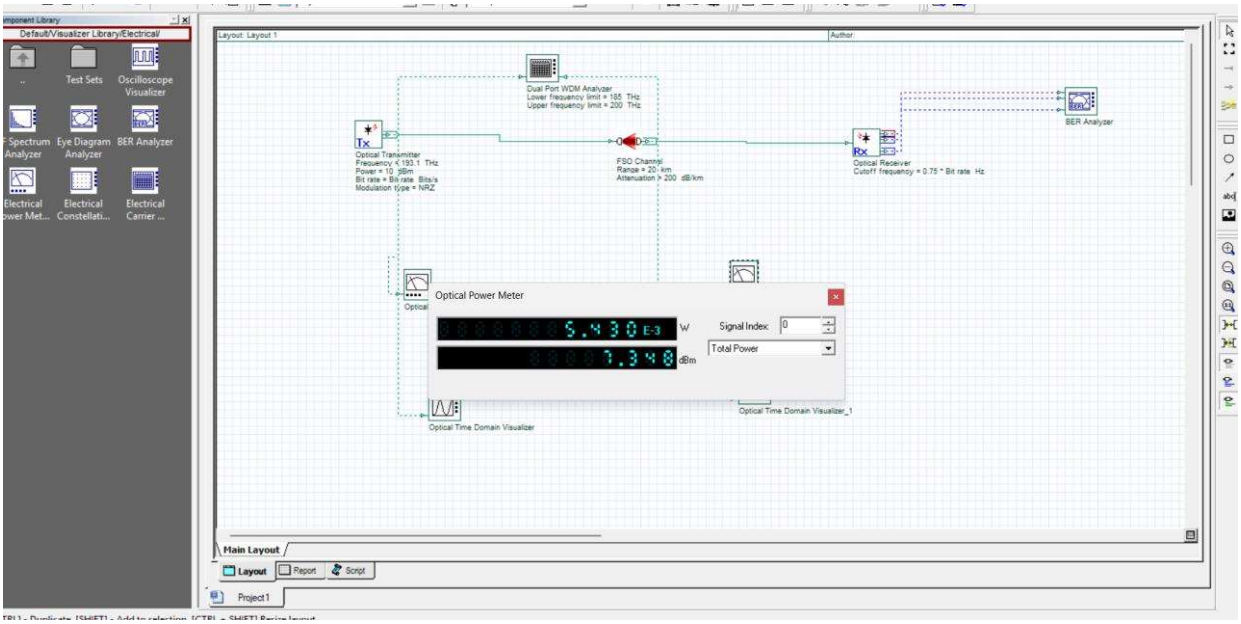
- Optisystem

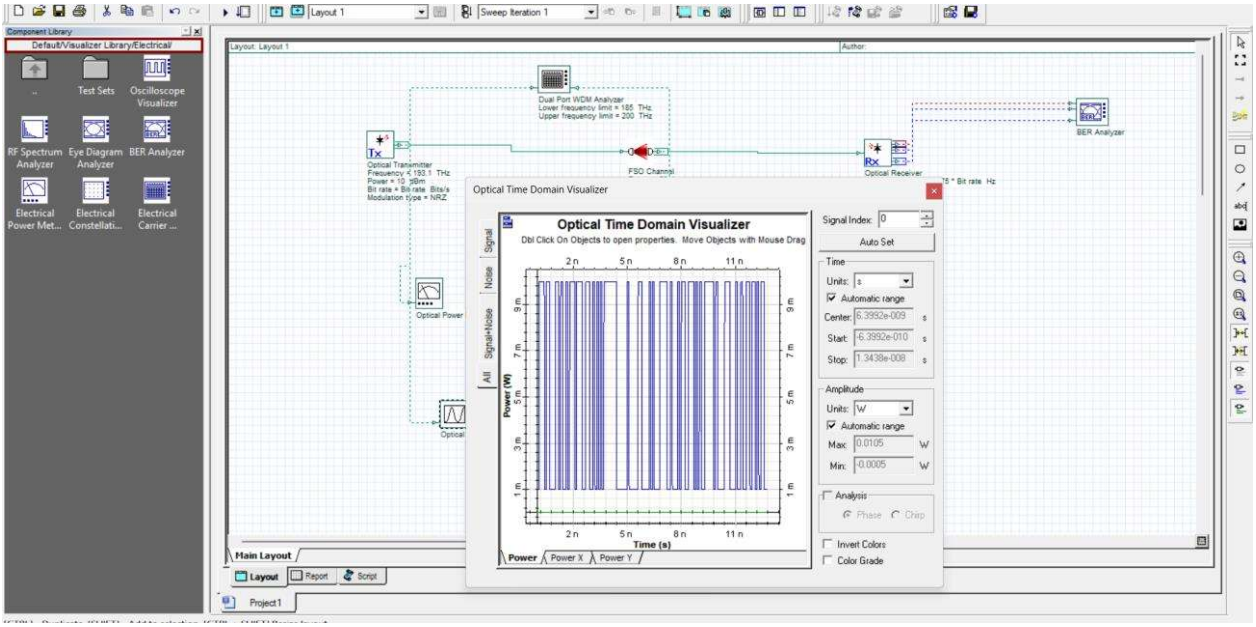
Circuit diagram:



Output:


BER Analyzer
Signal Index
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor
Min. BER
Eye Height
Threshold
Decision Inst.
Invert Colors
Color Grade
Patterns
Calculate Patterns
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Pattern 5
Q Factor
Min BER
Threshold
Height
BER Pattern
Main Layout
Layout
Report
Script
Project 1


Optical Power Meter
Signal Index
Total Power
Main Layout
Layout
Report
Script
Project 1



Reflection:

Thus, the performance of the Free-Space Optical (FSO) communication link under atmospheric turbulence was evaluated using OptiSystem software. The effects of atmospheric turbulence on the transmitted optical signal were studied by analyzing parameters such as received optical power, Q-factor, and bit error rate (BER). It was observed that increased atmospheric turbulence can cause signal fading and reduce link performance. Hence, the simulation helped in understanding the impact of atmospheric conditions on the reliability and quality of FSO communication systems.

Rubrics:

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
System Design & FSO Link Setup	Correct selection and connection of OptiSystem components for Free Space Optics link (Optical Transmitter, FSO Channel, PIN Photodetector, Low Pass Filter, Optical Power Meter, Time Domain Visualizer, BER Analyzer).	All optical transmitter, free space channel, receiver, and visualizer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the FSO link remains functional.	Component selection is partially complete; minor connection or configuration errors impede optical signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of physical input parameters: Wavelength (1550 nm), Modulation (OOK/NRZ), Transmit Power, FSO Range (1 to 15 km), and Atmospheric Attenuation (2 to 10 dB/km).	All input parameters and test cases across varying ranges and attenuation values are accurately set and executed.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple range or attenuation test cases are missing.	Incorrect input parameter settings or failure to run the required atmospheric test cases.	20%
Measurement & Power Monitoring	Accurate extraction and recording of received optical power, Q-Factor, BER, Eye Height, and Threshold across all range settings.	Precise data extraction across all FSO range cases, neatly tabulated with complete optical power readings and clear time-domain waveforms.	Data is collected with minor omissions in recording power meter readings, BER, or eye parameters.	Partial data collection with incomplete tables or missing optical power visualizer readings.	Inaccurate, incomplete, or missing data and measurements.	20%
Atmospheric & Link Loss Analysis	Detailed evaluation of geometric spreading loss, atmospheric attenuation, distance-dependent signal degradation, and power compensation feasibility.	Comprehensive analysis correctly relating range and attenuation variations to geometric loss, received power drop, eye height reduction, and bit error rate performance.	Clear analysis identifying basic trends in received power and eye height relative to FSO distance.	Basic attempt at result analysis without explaining physical mechanisms like atmospheric scattering or geometric loss.	Erroneous or missing analysis of simulation results and power meter outputs.	20%
Report & Conclusion	Identification of maximum reliable communication distance, power compensation trade-offs with diminishing returns, and structural clarity.	Report features clear time-domain waveforms, complete power tables, and insightful conclusions identifying optimal operational parameters under atmospheric impairments.	Satisfactory report with relevant visualizer displays and sound concluding summary.	Incomplete report with missing graphs, power readings, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

Result:

The simulation in OptiSystem provides a clear understanding of how FSO systems perform under varying ranges and atmospheric conditions. This analysis helps design robust FSO systems for different applications, ensuring reliable performance in various environments.